

Build Log

Print it, fill it in with a pen, keep it with the launcher

*One sheet, one build — the evidence that it
is sound*

This is the difference between a launcher you hope is fine and one you know is fine. Fill it in as you go, not afterwards from memory. Every blank on this page exists because somebody, somewhere, guessed instead of checking.

Builder _____
Adult in charge _____
Build started _____ **Finished** _____

Materials — copy from the printing on the pipe

Part	Nominal	Schedule / rating	Marking says (copy it)
Chamber pipe	_____	_____	_____
Barrel pipe	_____	_____	_____
Reducer	_____	_____	_____
Cleanout	_____	_____	_____

STOP HERE IF ANY LINE DOES NOT SAY PRESSURE-RATED

Every pressure-bearing part must read **Schedule 40** with an **NSF-PW** rating and a pressure figure. If a part says DWV, cellular core, foam core, or carries no pressure rating at all, it does not go in this build. That pipe fails by shattering rather than splitting, and no amount of care elsewhere compensates.

Cut list — measured twice, from one reference end

Piece	Target	1st measure	2nd	Square? (4 positions)
Chamber	_____	_____	_____	_____
Barrel	_____	_____	_____	_____

GOVERNED BY CRAFT 1 AND CRAFT 2

Every end deburred and chamfered

End	Inside	Outside	Checked by (initials)
Chamber, breech end	_____	_____	_____
Chamber, reducer end	_____	_____	_____
Barrel, reducer end	_____	_____	_____
Barrel, muzzle	_____	_____	_____

GOVERNED BY CRAFT 3

Joints — the page that matters most

For each joint: the date and time you welded it, whether primer was used, whether the squeeze-out bead was continuous all the way round with no gaps, and whether the witness mark stayed aligned.

Joint	Date & time	Primer	Bead all round	Witness mark aligned
Cleanout to chamber	_____	_____	_____	_____
Chamber to reducer	_____	_____	_____	_____
Reducer to barrel	_____	_____	_____	_____

GOVERNED BY CRAFT 4

THE CURE GATE — COMPUTE IT, DO NOT ESTIMATE IT

Latest weld finished at _____ (date and time).

Add 48 hours. **Earliest permitted first firing:** _____

Write that time here, and write it on the chamber in marker as well. Nobody fires this before then — not for a quick test, not because it feels solid, not because the weather is good today and bad on Sunday.

Igniter

Check	Result / initials
Holes round, no crazing, no cracks	_____
Electrode gap measured	_____ mm _____
Spark seen <i>and</i> heard, chamber open, before sealing	_____
Epoxy fillet continuous all round, both electrodes	_____
Epoxy fully cured before first pressure	_____

GOVERNED BY CRAFT 5

Pre-firing inspection — all ten, every session

Run the Craft 6 list. Initial each. Any single failure stops the session.

Check	Session 1	2	3
1 Cut faces square			
2 Edges deburred and chamfered			
3 Every weld bead continuous			
4 Cure time elapsed			
5 Penetrations sound, no crazing			
6 Chamber wall: no crazing or cracks			
7 Cleanout threads clean, seats fully			
8 Spark confirmed, chamber open			
9 Barrel bore clear (checked from breech)			
10 Pressure rating still legible on every part			
GOVERNED BY CRAFT 6			

Firing record

Date	Fuel dose	Angle	Range	Notes / anything unexpected
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Retirement. Any crazing, whitening, star crack or deformation of the chamber retires it permanently — cut it up so it cannot be reused and build a new one. Record the date and the reason here:

Why a form, and not just care

Because memory is confident and wrong. “I’m fairly sure we glued it Friday” is not a cure time, and the person saying it is not lying — they genuinely believe it. A written date removes the argument entirely, and it removes it in advance, which is when removing arguments is useful. Every serious workshop, laboratory and aircraft hangar in the world runs on exactly this idea.

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